

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

5

Code of Conduct:

I NETHRA P(192511012)

certify that this submission is my

original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

CO4 – ASSESSMENT TOOL 1

Course: Artificial Intelligence (CSA17)

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach, Target Variable, and Key Features

The healthcare dataset is a supervised inductive learning problem: the model generalises a decision rule from a labelled set of past patient records (input features → known diagnosis outcome) so that it can predict the outcome for new, unseen patients. Since the target column (diabetic / non-diabetic) is categorical, this is inductive learning for classification, and algorithms such as Decision Trees, Naïve Bayes, or Logistic Regression are suitable because they build an explicit, generalisable hypothesis from finite labelled examples (the classic inductive-bias framework).

Target variable: Diabetes_Status (binary: 1 = Diabetic, 0 = Non-Diabetic).

Key features and justification:

- Glucose Level — directly reflects blood-sugar regulation and is the single strongest clinical indicator of diabetes (elevated fasting/random glucose is a diagnostic criterion itself).
- BMI (Body Mass Index) — obesity is a well-established risk factor because excess adiposity increases insulin resistance.
- Family History — captures genetic predisposition; diabetes has a strong hereditary component, so a positive family history substantially raises prior risk.
- Age and Blood Pressure — supporting features: risk rises with age (Type-2 onset), and hypertension frequently co-occurs with metabolic syndrome, so they add incremental predictive value.

(b) Handling Class Imbalance

In a real patient population, non-diabetic cases greatly outnumber diabetic cases. A classifier trained naively on this data will be biased toward predicting the majority class, which is clinically dangerous because it maximises the chance of missing true diabetic patients (false negatives).

Recommended strategy: SMOTE (Synthetic Minority Over-sampling Technique). SMOTE generates synthetic minority-class (diabetic) samples by interpolating between a minority sample and its nearest minority neighbours in feature space, rather than merely duplicating existing rows.

Justification over alternatives:

- Vs. random under-sampling: under-sampling discards majority-class (non-diabetic) records, which throws away valuable information and shrinks an already limited medical dataset — undesirable when data collection is expensive.
- Vs. class weights alone: adjusting class weights is computationally cheap and can be combined with SMOTE, but by itself does not enrich the diversity of minority examples the model sees; it only re-weights the loss.
- SMOTE is preferred here (optionally combined with class weighting) because it increases minority-class representation without losing majority-class information, improving recall for diabetic cases — the clinically critical metric.

(c) Train-Test Split and Preprocessing

Split ratio: 70% train / 30% test. A 70:30 split is justified for a medical dataset of moderate size because it reserves a sufficiently large test set (30%) to obtain a statistically reliable estimate of generalisation performance — critical before any clinical deployment — while still leaving 70% for the model to learn reliable patterns. Stratified sampling is used on the split so that the diabetic:non-diabetic ratio is preserved in both sets, preventing the test set from becoming even more imbalanced than the source data.

Preprocessing steps applied:

- Normalisation (Min-Max or Z-score scaling) on continuous features — Glucose, BMI, Blood Pressure, Cholesterol, Age. Impact: brings all features to a comparable scale so that distance/gradient-based components (e.g., Logistic Regression, Naïve Bayes' Gaussian likelihoods) are not dominated by features with naturally larger numeric ranges (e.g., Cholesterol in the 100s–300s vs. BMI in the 15–40 range).
- Encoding of categorical fields — Family History (Yes/No) is label/one-hot encoded to 1/0. Impact: converts non-numeric data into a numeric form all the chosen algorithms can process, without introducing a false ordinal relationship.
- Missing-value imputation (median for skewed clinical variables such as glucose, mode for categorical fields) — Impact: prevents row-deletion from further shrinking the minority class and avoids bias from ad-hoc handling.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Construction — First Three Levels

Root node selection: Information Gain / Gini Index is computed for every candidate feature at the root. Illustrative values (based on typical predictive strength for this domain):

Feature	Gini Index	Information Gain
Glucose Level	0.21	0.38
BMI	0.29	0.27
Family History	0.33	0.19
Age	0.36	0.14
Blood Pressure	0.39	0.10

Glucose Level gives the lowest Gini Index (highest purity split) and the highest Information Gain, so it is selected as the root node — consistent with its known status as the strongest single clinical predictor.

Tree structure (first three levels)

Level 0 (Root): Glucose Level $\leq$ 140 mg/dL ?

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├─ Yes → Level 1: BMI  $\leq$  27 ?
│   └─ Yes → Level 2: Family History = No ?
│       └─ Yes → [Leaf: Non-Diabetic]
│       └─ No → Level 3: Age  $\leq$  45 ? (further split)
│   └─ No → Level 2: Age  $\leq$  50 ?
│       └─ Yes → [Leaf: Non-Diabetic]
│       └─ No → [Leaf: Diabetic]
└─ No → Level 1: Family History = Yes ?
    └─ Yes → [Leaf: Diabetic]
    └─ No → Level 2: BMI  $\leq$  30 ?
        └─ Yes → [Leaf: Borderline → further split on
Cholesterol]
        └─ No → [Leaf: Diabetic]
```

Reading: the tree first isolates patients by glucose level (the strongest single indicator), then refines the prediction using BMI and family history — the second- and third-ranked features — which mirrors clinical reasoning where glucose is the primary screening measure and BMI/heredity refine the risk estimate.

(b) Model Evaluation

Illustrative confusion matrix on the 30% held-out test set ($n = 300$):

	Predicted: Diabetic	Predicted: Non-Diabetic
Actual: Diabetic	72 (TP)	18 (FN)
Actual: Non-Diabetic	15 (FP)	195 (TN)

Metric	Formula	Value
Accuracy	$(TP+TN)/\text{Total}$	89.0%
Precision	$TP/(TP+FP)$	82.8%
Recall (Sensitivity)	$TP/(TP+FN)$	80.0%
F1-Score	$2 \cdot P \cdot R / (P+R)$	81.4%

False Negatives (18 cases): these are diabetic patients the model predicted as non-diabetic — the most clinically dangerous error, since these patients would receive no follow-up care.

Ways to reduce False Negatives, with clinical justification:

- Lower the classification decision threshold (e.g., flag as diabetic at predicted probability > 0.35 instead of 0.5) — trades some precision for higher recall, appropriate because missing a diabetic patient (FN) is clinically costlier than a false alarm (FP), which only triggers a confirmatory test.
- Apply class-weighted or cost-sensitive training so misclassifying the minority (diabetic) class is penalised more heavily during tree construction.
- Combine with SMOTE-balanced training data (Task 1b) so the tree does not under-represent diabetic-class splits.
- Add an ensemble layer (e.g., Random Forest) to reduce variance-driven misclassifications of borderline cases.

(c) Post-Pruning (Cost-Complexity Pruning)

Cost-complexity pruning (minimal cost-complexity / CCP) grows the full tree and then iteratively removes the weakest sub-tree — the one that adds least to overall accuracy relative to its complexity — controlled by the complexity parameter α . Increasing α prunes more aggressively.

Model	Tree Depth	Train Accuracy	Test Accuracy
Pre-Pruning (max depth unrestricted)	11	97.5%	84.0%
Post-Pruning (CCP, $\alpha = 0.015$)	5	91.0%	89.0%

The unpruned (pre-pruning) tree overfits: near-perfect training accuracy but a noticeably lower test accuracy, because it has memorised noise-level splits specific to the training records. The post-pruned tree sacrifices some training accuracy but generalises better (higher test accuracy) and is far shallower/simpler.

Recommendation for clinical deployment: the post-pruned model is more appropriate. Beyond better generalisation, a shallow tree (depth 5) is far more interpretable to clinicians, who need to see and trust the reasoning path (explainability is a regulatory and ethical requirement in healthcare AI), and it is less likely to make erratic predictions on new patients whose values fall outside patterns seen only a few times in training.

Task 3: Statistical Learning for Risk Stratification

(a) Naïve Bayes / Logistic Regression vs. Decision Tree

Logistic Regression is applied as the statistical learning model, since it directly outputs a calibrated probability of diabetes and is a standard clinical risk-scoring approach (analogous to established risk-score tools such as the Framingham score).

Model	Accuracy	AUC-ROC
Decision Tree (post-pruned)	89.0%	0.87
Logistic Regression	87.5%	0.91

Logistic Regression has slightly lower raw accuracy but a higher AUC-ROC, meaning it ranks/separates diabetic vs. non-diabetic patients more consistently across all decision thresholds, which is valuable for risk-stratification (e.g., assigning patients to low/medium/high-risk tiers) rather than a single hard classification.

When a statistical model is preferable:

- When the clinical use case needs a continuous, well-calibrated risk probability (risk stratification / triage) rather than a single yes/no label — Logistic Regression's sigmoid output is naturally suited to this.

- When the relationship between features and outcome is expected to be largely linear/monotonic (e.g., risk rises steadily with glucose) and the dataset is small-to-moderate, where Logistic Regression is less prone to overfitting than a deep tree.
- When regulatory bodies require a simple, auditable coefficient-based model (each feature's log-odds weight is directly interpretable).
- Decision Trees remain preferable when the true decision boundary is non-linear/rule-like (as clinical guidelines often are, e.g., 'if glucose > X AND BMI > Y') and when clinicians want an explicit, if-then explanation.

(b) Feature Importance Analysis

Rank	Decision Tree (Gini-based importance)	Logistic Regression (coefficient -based)
1	Glucose Level	Glucose Level
2	BMI	Family History
3	Family History	BMI

Both models agree that Glucose Level is the dominant predictor, and both place BMI and Family History as the second and third most important features — only their relative order (BMI vs. Family History) swaps between the two models. This is expected: the Decision Tree ranks importance by how much a feature reduces impurity across splits (favouring the feature with the cleanest numeric threshold, BMI), while Logistic Regression ranks by the magnitude of its linear coefficient (favouring the feature with the strongest linear/log-odds association, Family History). The overall agreement across both independent methods increases confidence that Glucose, BMI, and Family History are the true clinical drivers of the prediction, not an artefact of one algorithm.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

States (S): Discrete patient health stages derived from risk level, e.g., $S = \{\text{Critical, High-Risk, Moderate-Risk, Stable}\}$. Justification: treatment appropriateness depends on the patient's current physiological state, and a small set of clinically meaningful stages keeps the state space tractable while still capturing the health trajectory the agent must reason about.

Actions (A): $A = \{\text{Diet, Exercise, Medication, Monitor}\}$. Justification: these are exactly the interventions a clinician can prescribe, so the action space directly matches real-world decision options, making the learned policy actionable.

Reward function (R): $R = w_1 \cdot (\text{Health Improvement Score}) - w_2 \cdot (\text{Treatment Burden/Cost}) - w_3 \cdot (\text{Adverse Event Penalty})$. Justification: reward must encode not just whether the patient's health metrics (e.g., glucose, BMI) improved after the action, but also penalise unnecessary intervention (e.g., prescribing Medication for a Stable patient) and heavily penalise any action that leads to an adverse outcome, so the agent is not incentivised to over-treat.

Transition dynamics (P): $P(s'|s,a)$ — the probability a patient moves from state s to state s' after action a , estimated from historical longitudinal patient records (state transitions observed between consecutive check-ups). Justification: since exact physiological response to treatment is stochastic (patients respond differently), transitions must be modelled probabilistically rather than deterministically, which is exactly what the MDP framework is built for.

(b) Q-Learning: Table Updates Across 5 Episodes

Update rule: $Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$, with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.9$. Five patient episodes (Episode 1–5) are simulated; the table below shows the update for three representative state-action pairs across Iteration 1 (Episodes 1–2 average) and Iteration 2 (Episodes 3–5 average).

State-Action Pair	Q-value (Init)	Reward (r)	Iteration 1 Q-value	Iteration 2 Q-value
(High-Risk, Medication)	0.00	+8	0.80	1.42
(Moderate-Risk, Exercise)	0.00	+6	0.60	1.08
(Stable, Monitor)	0.00	+3	0.30	0.56

Sample calculation for (High-Risk, Medication), Iteration 1: $Q = 0.00 + 0.1[8 + 0.9 \cdot \max_{a'} Q(\text{High-Risk} \rightarrow \text{Moderate-Risk}, a') - 0.00] = 0.1 \times 8 = 0.80$ (assuming max next-state Q is still 0 in the first pass). In Iteration 2, the same pair is revisited after the agent has already learned some downstream values, so the bootstrapped term $\max_{a'} Q(s',a')$ is now non-zero, pushing the Q -value up to 1.42.

Convergence criterion used: training stops when the maximum absolute change in the Q -table between successive full passes over all state-action pairs falls below a threshold,

$|Q_{\text{new}}(s,a) - Q_{\text{old}}(s,a)| < \epsilon$ ($\epsilon = 0.01$) for all (s,a) , or after a fixed cap of episodes is reached — whichever occurs first. This indicates the Q-values have effectively stabilised and the greedy policy derived from the table is no longer changing.

(c) Final Learned Policy and Ethical Considerations

Extracted greedy policy $\pi(s) = \operatorname{argmax}_a Q(s,a)$ (illustrative, from the converged table):

State	Recommended Action (π^*)
Critical	Medication
High-Risk	Medication
Moderate-Risk	Exercise
Stable	Monitor

Ethical considerations of deploying an RL treatment-recommendation agent:

- Patient safety: the agent's policy must never be executed autonomously — it should function as a decision-support recommendation reviewed and approved by a licensed clinician before any action (especially Medication) is applied, since an RL agent trained on historical data can propose unsafe actions for patients whose profile differs from training data.
- Bias: if historical treatment/outcome data reflects unequal access to care across demographic groups, the learned reward signal and policy can inherit and amplify that bias (e.g., under-recommending Medication for a group that was historically under-treated). Regular fairness audits across sub-populations are required.
- Explainability: unlike the Decision Tree, a Q-table/RL policy can be opaque about why an action was chosen for a given state; clinicians and patients are entitled to a transparent rationale, so the deployed system should expose the contributing state features and expected reward alongside each recommendation.
- Consent and accountability: patients should be informed that AI is involved in generating (not making) treatment suggestions, and clear accountability must remain with the supervising clinician, not the algorithm.
- Reward misspecification risk: because the reward function is a proxy for true health outcomes, poorly designed rewards (e.g., over-weighting short-term glucose improvement) could encourage the agent to recommend aggressive interventions with

harmful long-term side effects — reward design itself requires clinical governance and ongoing monitoring after deployment.